Compliant Mechanism for Precise Maneuvering of Endotracheal Intubation Robot

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***Abstract*—The purpose of this project is to investigate the implementation of a compliant mechanism as a replacement for a segmented soft robotic chain for automated endotracheal intubation. The previous design utilized a series of rigid links actuated by tendons to achieve bending within the confined geometry of an endotracheal tube. However, this design introduced challenges in stiffness predictability and precise controllability. This work proposes an alternative solution by introducing a compliant mechanism with distal steering for smoother and more precise bending behavior. As part of this study, an analytical formulation will be derived using the pseudo-rigid body (PRB) method to relate tendon actuation with the tip deflection and curvature. Finite element analysis (FEA) will be used to validate the analytical predictions and evaluate stress and deformation characteristics. The results will assess the feasibility of compliant mechanisms for precision-guided intubation.**

I. Introduction

The Department of Biomedical Engineering at The Ohio State University had 22 capstone projects in the year 2025. Among these projects was an AI-assisted Endotracheal Intubation robot designed to guide and maneuver the endotracheal tube into a patient's trachea while adapting to physiological conditions in real time [1]. While this prototype demonstrated promising results, the team leading this project reported facing a variety of issues with the soft robot linkage design that controlled and maneuvered the endotracheal tube, lacking precise control of its endpoint due to the several points of deflection created by each link. This limitation serves as the main motivation for this work, seeking to provide a solution that offers increased control and predictability without sacrificing the flexibility required for navigating the airway through the use of a compliant mechanism.

II. Goals

- To develop the analytical formulation and the supporting equations to model the behavior of the mechanism using the PRB method.
- To develop the FEA model for the mechanism and use its results for validation.
- To develop an interactive computer application with a graphical user interface (GUI) that allows for the variation of key design and loading parameters.
- To compare the analytical and FEA-based results.
- To assess the efficacy of substituting the soft robot linkage with a compliant mechanism.

III. Problem Definition

The motion of the endotracheal intubation robot consists of a primary bulk motion that provides the driving force for inserting the endotracheal tube into the patient's trachea, followed by small adjustments made by the tendons of the soft robot linkage to guide the tube through the trachea while adapting to the patient's unique physiology.

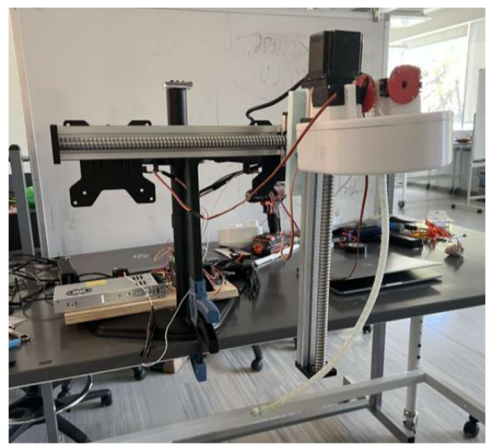

Fig. 1. Functional Prototype of Endotracheal Intubation Robot

The main problem arises from the nature of the intubation and the design of the intubation tube itself. The intubation tube has an outer diameter that fits securely within the patient's trachea and is designed to maintain its structural integrity within it to provide a stable pathway for air. As a result, the robot performing the intubation and micro adjustments has to balance dimensional, motion tracking, and stiffness constraints, being able to fit within the tube while controlling it precisely and bending it from within. This study focuses on implementing a compliant mechanism and varying its design parameters in order to evaluate their influence on desired controllability and stiffness.

References

[1] The Ohio State University, "Innovative AI-Assisted Intubation Project Shared at Capstone Showcase," *Department of Biomedical Engineering News*, May 2025. [Online]. Available: https://bme.osu.edu/news/2025/05/innovative-ai-assisted-intubation-project-shared-capstone-showcase.

[2] The Ohio State University, "Automated Endotracheal Intubation," *Ohio State Innovate*. [Online]. Available: https://innovate.osu.edu/available_technologies/62900/Automated-Endotracheal-Intubation.